AI Powered Study Tutor

A web app that gives students study notes matched to their level.

A student signs up and fills in a short assessment (age, grade, study time, and scores in Math, Science, English and Social). Machine learning models predict the student's level in each subject. When the student types any topic, an AI model writes notes for that topic at the right level.

How it works

1. **Register / log in.** Accounts and profiles are stored in SQLite.
2. **Assessment.** The student enters their details and subject scores. One XGBoost model per subject predicts an assessment score, which maps to a level: Beginner, Intermediate or Advanced.
3. **Study.** The student enters a topic, for example "Photosynthesis" or "Fractions".
    - The topic is sorted into Math, Science, English or Social by an LLM.
    - Notes are generated by an LLM (Qwen2.5-7B through the Hugging Face API) at the student's level in that subject.
    - The notes have an overview, core concepts, real-world use, and a short quiz. Math formulas are shown with MathJax.
    - Notes are cached on disk, so the same topic and level is not generated twice.
4. **Profile.** Shows past assessments and whether scores are going up or down.
5. **Study materials.** PDFs organised by subject, level and class.

Tech stack

- **Backend:** Python, Flask, Flask-Login, Flask-SQLAlchemy
- **Database:** SQLite
- **ML:** XGBoost, scikit-learn, pandas
- **AI notes:** Hugging Face Inference API (Qwen2.5-7B-Instruct)
- **Frontend:** HTML, Jinja templates, CSS, MathJax

Project structure

```
app.py                  Flask app: routes, login, notes generation, caching
model.py                Loads the saved models and predicts scores and levels
assess_score.py         Script used to build the assessment predictions
check_accuracy.py       Checks model accuracy
saved_models/           Trained XGBoost models (.pkl)
study_materials/        PDFs by subject, level and class
templates/              HTML pages
static/css/             Styles
```

Run it locally

```bash
git clone https://github.com/rupan-05/Ai_Powered_Study_tutor.git
cd Ai_Powered_Study_tutor

python -m venv venv
venv\Scripts\activate          # Windows
# source venv/bin/activate     # Mac/Linux

pip install flask flask-sqlalchemy flask-login pandas numpy scikit-learn xgboost requests
```

Set your free Hugging Face token (get one at huggingface.co/settings/tokens):

```bash
set HF_API_TOKEN=your_token_here          # Windows
# export HF_API_TOKEN=your_token_here     # Mac/Linux
```

Start the app:

```bash
python app.py
```

Open http://127.0.0.1:5000

Notes

- Never commit your API token. Keep it in an environment variable.
- The saved models predict from the details and scores the student enters. Model accuracy can be checked with `check_accuracy.py`.

Author

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